IoT-Sentry: A Cross-Layer-Based Intrusion Detection System in Standardized Internet of Things

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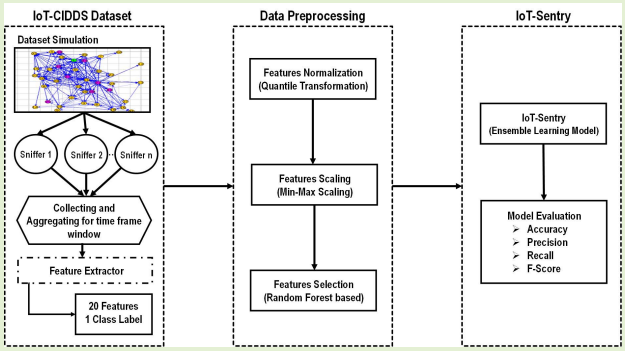
***Abstract*—With the widespread expansion of internet connected devices, securing the Internet of Things (IoT) has become one of the biggest challenges and a crucial issue for ensuring a secure and robust IoT vision. The usage of adhoc topologies and the resource constraints of IoT devices and networks makes first line of defense mechanisms like cryptography unsuitable to implement. Therefore, second line of defense mechanisms such as the intrusion detection system become fundamental to detect attacks. In this work, we propose IoT-Sentry, a cross-layer intrusion detection system to detect five different attacks with zero additional overhead. Also, there is a lack of rich, illustrative and concise public datasets in IoT for evaluation of intrusion detection models for IoT networks. Our research work fills this research gap by (1) developing a novel cross-layer IoT dataset which contains four simulation instances for the five attacks in static IoT networks with up to 100 nodes, (2) utilize novel features of cross-layer attacks, and (3) employ ensemble learning model to detect IoT attacks. In our work, the Cooja IoT simulator has been utilized for generating malicious and benign traffic. The IoT attack dataset is then analyzed and fed into centralized detection module implemented at the 6LoWPAN border router. On evaluation, IoT-Sentry achieves on average an accuracy of 99% for 4 out of 5 attacks. To the best of our knowledge, this is the first time that cross layer intrusion detection system catering to five attacks from different layers is developed for securing standardized IoT networks.**



***Index Terms*—Ensemble learning, cyber security, intrusion detection, cooja simulator, IoT dataset, 6LowPAN.**

I. Introduction

THE Internet of Things (IoT) is termed as the interconnection of devices including sensors and actuators running different software, supported by independent technology stacks and protocols. These devices often gather and share data obtained from the surrounding environment. According to a study by IoT Analytics, by 2025, the number of active connected IoT devices will be 30 billion each generating its own data [1]. Thus, the timely availability of authentic sensor data becomes crucial to ensure the success of these data-driven applications using constrained sensing devices. Without appropriate security mechanisms, these constrained device may be compromised, making it vulnerable to a number of attacks. For example, one of the largest Distributed-Denial-of-Service (DDoS) attacks was triggered using an IoT botnet in the year 2016. It was initiated with the Mirai malware [2]. Later in 2017, the U.S. Food and Drug Administration, in their study reported that cardiac devices exhibit serious vulnerabilities which, if exploited, could allow unauthorized access to the devices [3]. These attacks target the inherent vulnerabilities in IoT devices itself. Due to cut-throat competition amongst manufacturers, IoT devices are often rushed to the market with minimal or no security features [4]. Additionally, the adoption of 5G will accelerate the adoption of IoT applications in critical areas, and security mechanisms designed in the context of such applications can be complemented by intrusion detection security services provided by the infrastructure itself.

To handle the diverse communications in IoT and to create a set of protocols standard for communications over heterogeneous wireless communication technologies, along the goals of the Internet standard stack, a stack of standardized IoT protocols have been developed [5]– [6]. Throughout our dis-

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TABLE I
STANDARDIZED IOT PROTOCOL STACK

IoT Layer	IoT Protocol
Application Layer (Layer 6)	CoAP [10]
Transport Layer (Layer 5)	UDP [11]
Network and Routing Layer (Layer 4)	IPv6 [12], RPL [8], ICMPv6 [13]
Adaptation Layer (Layer 3)	6LoWPAN [9]
Data Link Layer (Layer 2)	IEEE 802.15.4 [7]
Physical Layer (Layer 1)	

cussion, we consider IoT as a low power wireless personal area network (LPWAN) with sensor nodes running the standardized protocol stack [5] and connected to a resource rich 6LoWPAN border router which provides the necessary interface between the Internet and the sensor devices. As shown in Table I, these standardized protocols have been developed for the low power and lossy wireless networks and include IEEE 802.15.4 [7], the Routing Protocol for low Power and Lossy Networks (RPL) [8], IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) [9], User Datagram Protocol (UDP) and Constrained Application Protocol (CoAP) [10]. The usage of standardized communication protocols is starting to enable critical application in areas where attacks can have dangerous consequences (e.g. industrial control, healthcare etc.)

There exists many attacks on IoT protocols aiming at the topology, the traffic and the resources of the network. A detailed taxonomy of attacks on RPL is given in [14]. Apart from the conventional communication security mechanisms like IPSec [15] and DTLS [16], RPL defines secure variants of all supported message types to ensure confidentiality, integrity and availability of routing information shared in the network. However, these security aspects rely on underlying layer 2 or layer 3 security algorithms. RFC 7416 [17] presents challenges of provisioning security aspects that may effect routing and which also need deliberations beyond the routing protocol. Being a stateless protocol, UDP suffers from two types of attacks, namely, the UDP reflection attack and the UDP flooding attack [18]. Since in IoT, all nodes are IPv6 addressable, it is worthwhile to mention the role of Neighbor Discovery Protocol (NDP) [19] in configuration of connections by: (i) determining link-layer addresses of neighbors; (ii) discovering routers willing to further forward the packets, and (iii) keeping track of node reachability. For this purpose, NDP utilizes ICMPv6 [13] packet types for router advertisement, router solicitation, neighbor solicitation, neighbor advertisement and redirect messages. The attacks on ICMPv6 include smurf and ping flood attacks which inundate the victim with ICMP requests without waiting for its response [20], [21]. Following are the restrictions of implementing first line of defense mechanisms like cryptography in securing IoT from attacks:

- Limited computational resources: IoT devices operate on low power, require less energy, have smaller memory and low computational capabilities and therefore are not fit to run conventional cryptographic mechanisms.
- Minimal or no security features: Due to cut-throat competition amongst manufacturers, IoT devices are often rushed to the market with minimal or no security features.
- Protection against internal attacks: IoT network is susceptible to internal attacks from authentic insiders and therefore, intrusion detection based on individual traffic analysis becomes critical.

Thus, there is a pressing need to secure IoT networks by incorporating security features at the network edge in IoT applications using constrained sensing devices to protect against disruption of communications, availability of data, battery depletion attacks etc. [22]. Hence, the usage of a second line of defense, particularly Intrusion Detection Systems (IDSs), is deemed important for detecting malicious intrusions or attacks. Kamaldeep *et al.* [22] provides a comprehensive taxonomy of the various types of IDS to secure the IoT. Most of the state-of-the art intrusion detection solutions in IoT focus only on routing layer attacks and lack evaluation on larger testbeds with multi-attacker scenarios.

At this end, we propose IoT-Sentry,[1] a cross layer intrusion detection system (CIDS) to detect cross layer attacks in standardized IoT. To handle vast traffic generated in IoT environment, we make use of machine learning (ML) algorithms to perform traffic analysis. In order to develop such ML based intrusion detection solutions that identify and detect intrusions, the availability of a realistic dataset is indispensable [23]. Given the novelty of IoT-Sentry and its use of packet inspection technique for cross layer features, we develop a novel dataset with cross layer attacks on RPL, UDP and ICMPv6 protocol in the standardized IoT stack. We call it the cross layer intrusion detection dataset for IoT (IoT-CIDDS).[2]

The major contributions of our work are as follows:

- A novel cross layer network attacks dataset for IoT, IoT-CIDDS containing malicious and normal traffic traces from multiple layers of the IoT communication stack. To the best of our knowledge, our proposed dataset represents the first comprehensive dataset of cross layer attacks in IoT.
- An analytical study of the developed dataset which includes data normalization, feature transformations and feature correlations study for intrusion detection algorithms.
- A lightweight intrusion detection framework, IoT-Sentry developed to detect cross layer attacks in IoT based on ensemble based machine learning.
- Demonstration of the applicability of IoT-Sentry on proposed dataset and evaluation of its performance. IoT-Sentry achieves an accuracy of upto 99.5% and F-score (up to 99%) (refer to Section V for more details).

The rest of the paper is organized as follows: Section II reviews relevant literature in this area. In Section 3, we propose our novel cross layer intrusion detection system for IoT including the description and generation of the developed dataset. Section 4 gives insights into the results and their

[1] Sentry means a soldier stationed to keep guard or to control access to a place, as proposed IDS aims to guard attacks against IoT.
[2] Available at: https://github.com/manishamalik53/IoT-CIDDS

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analysis. Finally, Section 5 concludes our work followed by scope for future work.

II. Background and Related Work

In this section, we review the available proposals on IDS in IoT, present their limitations and discuss the main motivations for our proposal. Throughout the review process, we discuss proposals based on the standardized protocols like RPL on routing layer, ICMPv6 on networking layer and UDP on transport layer. Existing work on intrusion detection for the conventional Internet belongs to two main classes: rule or signature based and anomaly based. Signature based IDS detects known intrusions based on a database of rules or signatures. Anomaly based IDS, on the other hand, detect unknown intrusions by looking for anomalies in communications or actions when compared with the expected or normal behaviour. Anomaly based IDS may be based on statistical, data mining or machine learning (ML) methods. A detailed and comprehensive taxonomy of IDS as second line of defense mechanisms for securing the IoT is given in [22] where apart from the conventional IDS classification, the authors emphasized that traditional IDSs are unsuitable for low power and lossy IoT networks and hence, the researchers must acclimate their attention in developing lightweight IDS specifically targeting the IoT networks.

In [32], the authors developed a hybrid IDS by keeping a check on rank and node IDs for the detection of anomalies. However, the work was evaluated in a closed setup with few nodes and the accuracy decreased as network was made lossy and the number of nodes increased. Whilst many researchers followed their footsteps, the works on evaluation of machine learning (ML) based IDS using IoT relevant dataset are less numerous [24]–[26]. Such works are significant to allow continuous monitoring and analysis of the large amount of data generated by the realistic IoT devices.

To begin with, the works in [24] implemented deep learning based detection of Decreased Rank (DR), Hello Flood (HF) and Version Number (VN) modification attack with Artificial Neural Networks (ANN). Their algorithm detected HF attack with 99.5% accuracy, VN attack with 95.2% accuracy, and DR attack with 94.9% accuracy. However, the authors proposed three different neural network models for the three attacks using different set of features which eventually increases the training time. Napiah *et al.* [25] utilized the Compression Header of 6LoWPAN to detect WM, SN, and HF attacks and their combinations thereof. The authors captured and analyzed the data from 6LoWPAN compression header and extracted features using Best First search and Greedy stepwise algorithms. However, the simulation in Weka was run for a small duration of 30 minutes for each attack with only eight nodes. Hence, the dataset was considerably small resulting in fewer samples for training and testing. The authors in [26] implemented the Kernel Density Estimation algorithm to detect VN, HF, and BK attacks. They evaluated the system by using features such as the count of RPL control messages (DIO, DIS and DAO), number of DODAG versions, and the ratio of transmitted to received UDP packets. In another research work in [27], genetic programming was used to detect intrusions in a network of 26 nodes with ten percent of the nodes being attacker nodes. However, the simulation was run for a small duration of 20 minutes with a smaller set of nodes.

Moreover, the works in [28] put forward the concept of Self-Organizing Map (SOM) neural network to cluster routing attack on RPL. Nonetheless, the performance of the IDS was not validated statistically. The authors in [30] developed a comprehensive Keeper dataset [33] by collecting real-world data from devices like thermostat, babycam, nestcam, wemo switch, and dlink sirens. They proposed IoT-KEEPER, an edge system for online traffic analysis using fuzzy C means clustering and fuzzy interpolation scheme to detect Mirai, ARP poisoning and DoS attack with 99% accuracy.

Developed in 2019, ensemble learning based NIDS (ELNIDS) [29] protected against seven RPL routing attacks. This study included four different classifiers but the accuracy of detection for each attack type was not specified. Recently, the authors in [31] claimed to develop a cross layer IDS for RPL-based IoT where they extracted features from the link layer and the network layer. However, the attacks were only restricted to the networking layer targeting the routing protocol RPL with the hello flood, worst parent, and the version number attacks. The authors in [20] propose the sole analytical intrusion detection framework to detect ping of death attacks in IoT networks. The works in [21] survey machine learning techniques to detect ICMPv6 based DDoS attacks and highlighted that there is a need to develop ML-based solution in IoT detect ICMPv6 based attacks.

The state-of-the art proposals closely related to this proposed framework is compared in Table II. The different characteristics of each intrusion detection system, such as detection mechanism, attack type, the size of the network and if it was evaluated on simulated dataset. Although they tend to address certain challenges, these proposals present the following drawbacks:

1) *Lack of cross layer intrusion detection dataset:* Existing IDSs are based on datasets that focus only on routing layer attacks on RPL and none of them cater to well known attacks on other layers of the IoT stack.
2) *Lack of multi-attacker scenarios:* Most of the proposals only consider a single attacker scenario whereas a realistic IoT network may be compromised by more than one attacker at a time. The evaluations have been performed in a limited way with a single malicious node. Hence, the impact of multiple attackers on the network was skipped during the evaluation of proposed solutions.
3) *Evaluations on small IoT networks:* The scale of IoT networks on which these proposals have been tested are comparatively small.
4) *Additional Overhead on IoT devices:* Most of existing proposals introduce communication overhead on IoT devices, making them inefficient for resource constrained IoT networks.

Therefore, we propose cross layer intrusion detection system for IoT (i.e. IoT-Sentry) capable of detecting attacks on RPL as well as on ICMPv6 and UDP. We implement five attacks, namely, the UDP Flooding, Blackhole, Selective Forwarding,

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TABLE II
REVIEW OF RELATED WORK

Ref.	Year	Detection Mechanism	Dataset Type	Attack Type	IoT Layer and Protocols	Number of Nodes	Number of Extracted Features	Data Type/ Format	Performance Metrics
[24]	2018	Deep Learning	Simulated	DR, HF and VN	Routing Layer-RPL Protocol	10-100	18	CSV	Accuracy - 99.5% (HF), 95.2% (VN), 94.9% (DR)
[25]	2018	Machine Learning	Simulated	WM, SN and HF	Routing Layer-RPL Protocol	8	5	PCAP	99.4% TPR
[26]	2019	Kernel Density Estimation	C Implementation	HF, VN, and BK	Routing Layer-RPL Protocol	12	5	-	90%TPR (HF), 96%TPR (VN) and 0.5% FPR (Average)
[27]	2019	Genetic Programming	Simulated	HF and VN	Routing Layer-RPL Protocol	26	50	-	96.08% (HF), 97.97% (VN)
[28]	2019	Self Organizing Maps	Simulated	HF, SN, and VN	Routing Layer-RPL Protocol	-	6	CSV	-
[29]	2019	Ensemble Learning	Simulated	SN, BK, Sybil, Clone ID, SF, HF and Local Repair attacks	Routing Layer-RPL Protocol	-	20	CSV	Accuracy - 94.07%
[30]	2020	C Means Clustering	Real World Testbed	Mirai, ARP Poisoning and DoS	-	10	10	PCAP	Accuracy - 0.99
[31]	2020	Neural Network	Simulated	VN, WP, and HF	Routing Layer-RPL Protocol	51	31	CSV	Detection Rate (96.88%)

Note: HF-Hello Flood, VN- Version Number, WM- Wormhole, SN- Sinkhole, ARP- Address Resolution Protocol, DR- Decreased Rank, WP- Worst Parent, PCAP-Packet Capture, CSV- Comma Separated Value, TPR- True Positive Rate, FPR- False Positive Rate, BK-Blackhole, DoS- Denial of Service - : Represents missing attribute

Hello Flooding attack and the ICMPv6 Flood attack. It represents a breakthrough increment over current proposals. IoT-Sentry uses traffic sniffing approach and incurs no additional overhead on the deployed IoT devices. This makes IoT-Sentry extremely lightweight and applicable for battery powered IoT networks. In addition to that, we propose a novel dataset to evaluate the performance of proposed IDS.

III. DESCRIPTION OF A CROSS-LAYER INTRUSION DETECTION DATASET IN IOT (IOT-CIDDS)

In this section, we present our novel CIDDS for IoT. Although the conventional datasets like the KDD'99 [40] have been widely used for evaluating IDS, their aptness in revealing real characteristics of IoT networks has been heavily criticized in [22]. The main criticisms include lack of representative data implementing IoT protocols and thus absence of IoT attacks in the dataset. Hence, it becomes infeasible to utilize such dataset for intrusion detection in IoT. Table III summarizes the characteristics of all these datasets. We found that well known DARPA, KDD'99 [40] or other similar datasets developed for wired networks will not result in optimum creation of IDS for the IoT ecosystem. Other newer developed datasets for IoT like UNSW-NB15 [34], RPL-NIDS [36], IoT-Botnet present several challenges, for example, representing only particular scenarios like botnet malware [37] or smart cities network [39], focusing only on routing attacks [24], [36], [38], lack of cross layer attacks on standard IoT protocols and non-availability in public domain for further research.

As is evident from the literature, no study caters to attack detection on cross layer IoT dataset based on both transport layer and network layer activities. To enhance the richness and comprehension, host based features such as the power consumption can also be used for detection, based on data retrieved using the Powertrace utility. Also, most of the IDS focus on detecting a single attack at a time whereas in reality while defending the network from single attack, another type of attack can put the network at risk. Additionally, as the number of devices and attackers increase, various kinds of dynamic responses may be exhibited by the network. Our objective is to fill this gap by proposing IoT-CIDDS.[3] We target five attacks across layers of the stack, namely, the UDP Flooding, Blackhole, Selective Forwarding, Hello Flooding attack and the ICMPv6 Ping of Death attack. All these attacks affect one or more features of the normal operations of the network. For instance, the blackhole and selective forwarding attacks cause packet loss and UDP and hello flooding attacks increase the number of specific control packets in the network. The dataset is novel in terms of attacks considered and traffic feature extraction by analyzing the packets. We use Cooja [41] to implement the five attacks (four simulations for each attack). For every simulation scenario, Cooja stores a Packet Capture (PCAP) file containing complete information of packets transmitted during the simulation. Table IV presents the number of packets saved during the simulations of attacks.

[3] Available at: https://github.com/manishamalik53/IoT-CIDDS

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TABLE III
COMPARISON OF IOT APPLICABLE DATASETS

Datasets	Reference	Features	IoT Traces	IoT Layer and Protocols	Number of Simulated IoT Nodes	Labeled Dataset	Flow Analysis/Deep Packet Inspection	Attack Type	Format	No. of Features	Year
(+) UNSW-NB15	[34]	Network	×	×	×	True	Flow	Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Recon-naissance, Shellcode and Worms	CSV	49	2015
(-) Sivanathan	[35]	Network	✓	Application Layer-HTTP	21	True	Flow	×	CSV	12	2017
(-) RPL-NIDDS17	[36]	Network	✓	Routing Layer-RPL Protocol	×	True	Flow	Clone ID, HF, Local Repair, SF, SN, BK, Sybil	CSV	20	2019
(+) Bezerra	[37]	Host	✓	Botnet	3	×	×	Mirai and Bashlite, Hajime, Aidra, Tsunami and Dofloo	CSV	4	2018
(+) IoT Routing Attack Dataset (IRAD)	[24]	Network	✓	Routing Layer-RPL Protocol	100	True	Deep Packet Inspection	DR,VN,HF	CSV	18	2018
(+) RADAR	[38]	Network	✓	Routing Layer-RPL Protocol	16	False	Deep Packet Inspection	BK, SF, SN, HF, Clone ID, Sybil, WM, VN, Replay, DIS, Local Repair	CSV	11	2021
(-) Smart Home Dataset	[39]	Network	✓	Transport Layer-TCP	4	-	Flow	IoT Malware	-	38	2021
(+) Proposed	**-**	**Network**	✓	**Transport, Network and Routing Layer-UDP, RPL and ICMPv6 Protocol**	**100**	**True**	**Deep Packet Inspection**	**UDP Flooding, BK, SF, HF, ICMPv6 Flooding**	**CSV**	**21**	**-**

(+) Indicates that the Dataset is publicly available. (-) Indicates that the Dataset is not publicly available. 1. Host based dataset. 2. Only DDoS attack Note: HF-Hello Flood, VN- Version Number, WM- Wormhole, SN- Sinkhole, ARP- Address Resolution Protocol, DR- Decreased Rank, WP- Worst Parent, PCAP-Packet Capture, CSV- Comma Separated Value, TPR- True Positive Rate, FPR- False Positive Rate, BK-Blackhole, DoS- Denial of Service
× : Represents missing or invalid attribute

It is possible to notice that we not only captured packets on individual attacks but also scenario where simultaneously multiple attackers are targeting the IoT network.

All the features examined by Cooja are extracted from un-encrypted network traffic. The unsecured mode of RPL and other protocols is usually deployed to comply with lightweight requirements of IoT devices because for scalable security, RPL requires public key encryption, and we can realistically consider that in most scenarios devices employed may not be able to support it. Even if secured version is implemented, IoT-CIDDS's features can be extracted by having the knowledge of the security keys, assuming that our system has full access to keys since it serves to secure the network. The PCAP files obtained from Cooja is similar to traffic obtained from specially deployed sniffer devices in the network which serve the centralized IDS. To simulate each attack, we ran actual code on Zolertia [42] sensor motes in the Cooja IoT simulator. We ran Contiki inside a virtual machine with 4 VPUs and 10 GB RAM. We ran the simulations from 10 to 100 nodes without the support of any additional infrastructure. The network topology considered for data collection and consequently for detection is given in Figure 1. This sample topology consists of 1 sink node, 4 attacker nodes and 15 benign nodes. The sink node is connected to the border router by SLIP (Serial Line Internet Protocol) serial line utility which is further connected to the Internet via Ethernet. The benign and attacker nodes communicate with sink node by RPL messages. The sniffer nodes capture the network traffic and send the captured traffic to the 6LoWPAN border router through wireless link. The simulation and network parameters are listed in Table V.

IV. PROPOSED CROSS-LAYER INTRUSION DETECTION SYSTEM FOR IOT: IOT-SENTRY

In this section, we propose IoT-Sentry, a novel cross-layer intrusion detection system for standardized IoT networks. We begin with an overview of the proposed intrusion detection system in Section IV-A, followed by a detailed description of IoT-Sentry's architecture and framework: *traffic collection* in Section IV-B, *feature extraction and selection* in Section IV-C and *intrusion detection* in Section IV-D.

A. Overview

Recall that IoT networks are lossy and resource constrained which use IPv6 as the network and RPL as routing protocol.

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TABLE IV
DETAILS OF ATTACKS IN CROSS-LAYER IoT DATASET

		Malicious			Benign	
	Attack	Nodes	Malicious/Benign Nodes	Total Packets	Nodes	Total Packets
Transport Layer	**UDP Flooding**	10	2/8	53014	10	53006
		20	4/16	53038	20	53031
		40	8/32	52853	40	54081
		100	10/90	16429	100	16118
Routing Layer	**Selective Forwarding**	10	2/8	74259	10	53006
		20	4/16	50568	20	53031
		40	8/32	50123	40	54081
		100	10/90	16096	100	16118
	Blackhole	10	2/8	69456	10	53006
		20	4/16	53169	20	53031
		40	8/32	40362	40	54081
		100	10/90	16589	100	16118
	DISFlooding	10	2/8	53041	10	53006
		20	4/16	53650	20	53031
		40	8/32	52239	40	54081
		100	10/90	14589	100	16118
Network Layer	**ICMPv6 Flooding**	10	2/8	57245	10	53006
		20	4/16	63108	20	53031
		40	8/32	52626	40	54081
		100	10/90	15478	100	16118

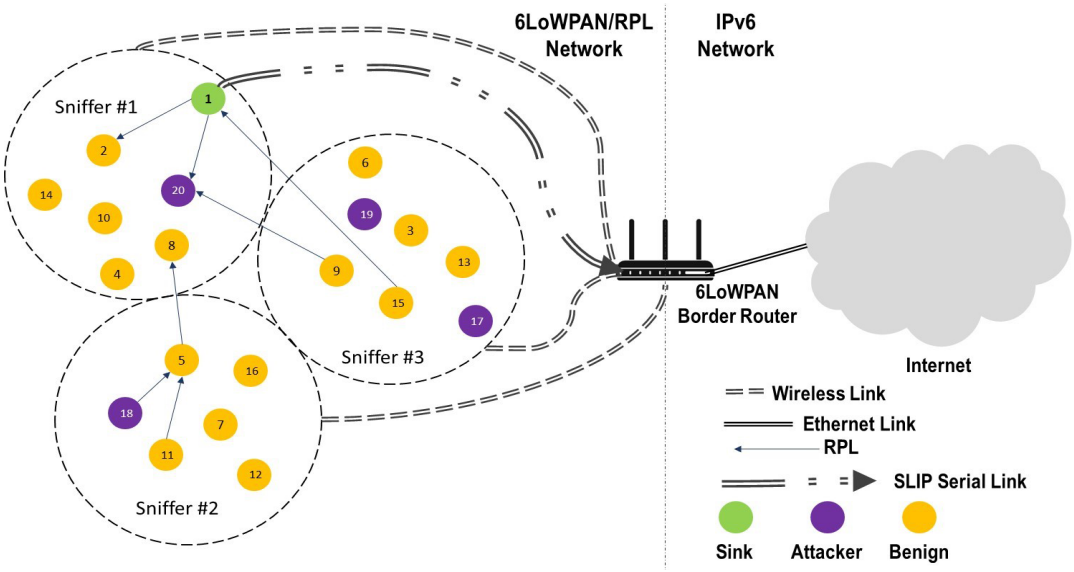

Fig. 1. Network topology of IoT-sentry: architecture and placement.

TABLE V
NETWORK AND SIMULATION PARAMETERS

Parameter	Value
Number of Nodes	10-100
Sink Node	1
Radio Environment	Unit Disk Graph Medium (UDGM)
Routing Protocol	RPL
MAC Protocol/Adaptation Layer	CSMA/6LoWPAN
Simulation Time	24 hrs
Sensor Node Type	Zolertia Z1

The ability of a protocol or system to be deployed in such constrained environment is one of the design requirements of any system for IoT. Some of these design requirements are:

- *R1 - Lightweight with no Additional Overhead:* The proposed system should not add any communication and computational overhead on the IoT devices and networks. Since RPL already exchanges a number of control messages, the desired system should not use RPL messages for information gathering since such an approach could adversely affect the routing performance of the network.
- *R2 - Attack Resistant:* The proposed system should be able to detect a wide range of attacks.
- *R3 - Scalability:* The proposed system should be scalable and work with large number of nodes.
- *R4 - Ease of Integration:* Easy integration of detection system in existing IoT networks.

IoT-Sentry satisfies all these requirements where newer attacks on a large number of nodes can be launched and detected easily. Also, it follows the concept of "Plug and play" and thus it is easier to integrate with existing IoT networks running the standardized protocol stack. Keeping in view the design requirement R1, we use a centralized approach and place IDS modules in 6LoWPAN border router. Also, contrary to other proposals, we do not use RPL control messages to gather information, since such an approach could cause an impact on the performance of the system. As shown in

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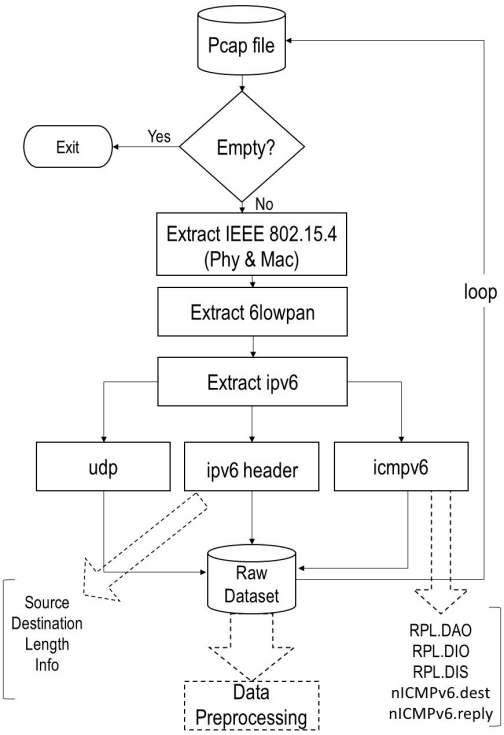

Fig. 2. Data collection flowchart.

Figure 1, the sniffer node gathers traffic information of the network and sends it to centralized module for analysis and intrusion detection, as we describe in Section IV-D.

B. Traffic Collection

For traffic collection, IoT-Sentry uses passive sniffing devices which may be optimally placed according to [43], [44]. The gathered traffic is in the form of PCAP file which is then sent to the 6LoWPAN border router over some other secure channel. In realistic implementations, it can be some out-of-band (OOB) mechanism like secure WiFi link between the sniffer nodes and 6LoWPAN. A review of such OOB based authentication mechanisms in IoT is given in [45]. The details of communication between 6LoWPAN border router and sniffers is out of the scope of this paper. The PCAP file contains information like the source IP, destination IP, length of the packet, the communication protocol, and related information (e.g. DIO, DIS, DAO, Data, etc.). By using the packet analysis tool, Wireshark [46], we then convert the PCAP file to Comma Separated Value (CSV) for preprocessing and feature extraction. The CSV files obtained from the PCAP conversion is our raw dataset. Figure 4 shows the process involved in extracting features from PCAP file captured from the IoT network. A sample of the raw dataset is shown in Table VI and the extracted features are shown in Table VII.

C. Feature Extraction, Scaling and Selection

IoT-Sentry monitors node's activities by collecting traffic and extracts features as described in Table VII. We consider the approach discussed in [24] where, we extracted 21 features by fragmenting the dataset on time window of 1000 ms rather

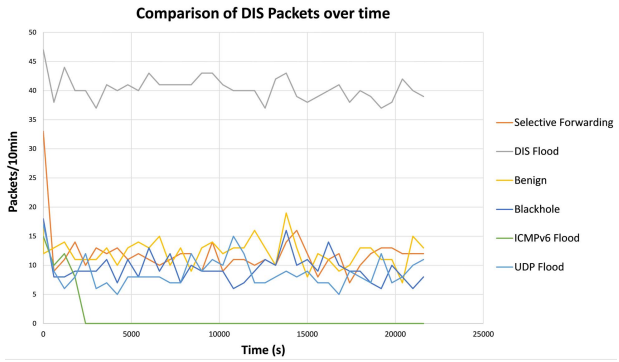

Fig. 3. DIS packets comparison.

than using global statistics which might badly affect the weight calculation of features. The attacks were implemented by compromising a node and modifying the code in Contiki OS. For UDP flooding attack, attacker floods the victim by sending packets on random ports. The victim, in return, generates a number of ICMPv6 destination unreachable messages making it unavailable to valid requests. In case of selective forwarding attack, the number of data packets decrease in the network. Similarly, the blackhole attack affects the overall throughput of the network. The number of sent packets increase as compared to received for nodes which are en-route from blackhole attacker to the root or the sink. Likewise, in DIS flooding attack, the network is flooded with solicitation packets irrespective of the received DIO messages.

Thus, we can say that when any of the target attack occurs, the number of certain type of packets increase in the network and the ratio of transmitted to received packets is also affected. Figure 3 shows the DIS messages from each scenario, and as can be seen, the DIS flooding attack scenario has the maximum number of sent DIS messages. However, when we compare the statistics of blackhole and selective forwarding, it is evident that number of DIS packets will be much lower in case of blackhole as attacker drops all the packets in that case. On the other hand, the number of ICMPv6 Destination Unreachable messages increase when UDP flooding attack occurs and number of received packets increase when this attack is initiated. Such features can be used as anomalies to detect intrusions in the network. Similarly, the count of DIO, DAS, DAO and Data packets are also calculated. Additional statistical features include the transmission and reception rate of the nodes, source IP, destination IP, packet length and simulation time.

In IoT, the labelling of dataset based on the flow is not feasible due to the huge amount of data generated from nodes. Therefore, to address this issue, we follow the approach implemented in [24], [47] and label the data based on simulation vector. Thus, the simulation vector which contains malicious node and activities is labelled as malicious or 1 and benign traffic is labelled as 0. To enhance the learning performance of IoT-Sentry, the features are normalized by applying *Quantile Transformation*, which distributes feature values to normal distribution. The feature values are later scaled with *Min-Max scaling* to 0-1 range for better classification performance. Since IoT-CIDDS is produced by processing network

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TABLE VI
A SAMPLE OF RAW NETWORK TRAFFIC DATASET

Number	Time	Source	Destination	Protocol	Length	Info
356	161.673	fe80::c30c:0:0:4	ff02::1a	ICMPv6	97	RPL Control (DODAG Information Object)
357	165.677	fe80::c30c:0:0:3	fe80::c30c:0:0:7	ICMPv6	102	RPL Control (DODAG Information Object)
358	167.193	::c30c:0:0:1	::c30c:0:0:4	UDP	48	Data (5678 → 8765 Len=7)
359	169.066	::c30c:0:0:7	::c30c:0:0:1	ICMPv6	102	RPL Control (Destination Advertisement Object)

TABLE VII
EXTRACTED FEATURES FROM RAW NETWORK TRAFFIC DATASET

Sr.No.	Feature Name	Feature Description
1.	Time	Simulation Time
2.	Source	IP address of source node
3.	Destination	IP address of destination node
4.	Length	Length of the packet
5.	Info	Packet Information
6.	TR	Transmission Rate per second of each node
7.	RR	Reception Rate per second of each node
8.	TR/RR	Ratio of Transmission to Reception Rate
9.	nTP	Number of transmitted packets
10.	nRP	Number of received packets
11.	ATT	Average Transmission Time
12.	ART	Average Reception Time
13.	TTT	Total Transmission Time
14.	TRT	Total Reception Time
15.	nDAO	Number of DAO packets
16.	nDIS	Number of DIS packets
17.	nDIO	Number of DIO packets
18.	nData	Number of Data packets
19.	nICMPv6.dest	Number of ICMPv6 Destination Unreachable messages
20.	nICMPv6.reply	Number of ICMPv6 reply messages
21.	Label	Malicious/Benign

packets captured from a simulated network in PCAP form, correct modeling of the effects of the attack on the network becomes crucial. As is depicted in Figure 4, feature selection in IoT-Sentry is performed with random decision trees, the Random Forest. The idea of using randomized decision trees lies in their ability to balance noisy and unbiased model and create a model with lower variance [24]. The tree-based strategies used by random forest creates a subset of most important features. Another advantage of using Random Forest is their ability to handle high dimensional data and large training examples. For this purpose, we implemented the *RandomForestClassifier* function from *sklearn* [48] library. Clearly, the feature importance will change for each type of attack as importance rates are based on the dataset values. The most important and least important features are dropped to avoid over-fitting and under-fitting respectively. The results are evaluated by executing the learning algorithm. This process is iterated until the best subset of features is obtained.

D. Intrusion Detection and Classification

IoT-Sentry detection modules are based on Ensemble Learning (EL), a supervised ML technique combining several set of ML models to improve the predictive power, robustness and stability of the model. The combination of individual ML models tend to make ensemble learning less data sensitive and more flexible. In addition, the use of bagging models in EL avoids overfitting. With bagging as ensemble method and decision trees as individual model, Random Forest (RF) model is one of the most common model in ensemble learning. RF has the capability to generate reasonable predictions across a wide range of data with minimum configuration. To make predictions on overall class of the data, RF takes the average of each tree prediction through a process called as bootstrap aggregating [49]. The authors in [50] compared RF model to other models for intrusion detection and concluded that RF obtained better precision, recall, accuracy and F1 score.

The proposed CIDDS dataset has been used to train and test the RF classifier and results have been compared. Since ours is a simulated dataset, we have run extra experiments with other classification models also with an aim to baseline our model for the developed dataset. The results are given in Table X.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Environment and Parameter Setting

In our previous work in [22], we discuss and classify standard validation techniques to prove the validity of any IDS in IoT. These validation techniques include: experimental, hypothetical, theoretical and simulation. In this paper, we follow the simulation approach to determine the efficiency of IoT-Sentry. All simulations were conducted on system with following specifications:

- Dataset Generation: Cooja Simulator [41] running on Contiki OS [51] in VirtualBox with 4 VPUs and 10 GB RAM. The simulations runs from 10 to 100 nodes without the support of any additional infrastructure for duration of 24 hours.
- Prototype Implementation: Our IoT-Sentry implementation involves a centralized machine connected to the sniffing devices through a secure channel. This centralized machine runs modules for traffic assembling, feature generation, extraction and selection. The feature engineering and ensemble learning based anomaly detection are implemented using Python libraries [48].

B. Intrusion Detection Performance Metrics

The standard performance metrics to evaluate and assess the performance of IDS are classified into resource utilization and detection related metrics [22]. The resource utilization metrics measure whether the IDS has any detrimental effect on the performance of the devices and network. These metrics include CPU utilization, memory overhead and energy consumption. As explained in Section IV.B, IoT-Sentry passively sniffs for traffic and sends the captured traffic to border router via secure OOB mechanism. Thus, we deduce that IoT-Sentry is lightweight and adds no additional resource overhead on the IoT devices and the network.

The detection related metrics [22] used for evaluation are accuracy, precision, recall and F-score. All these metrics are derived from confusion matrix given in Table VIII.

- *Accuracy:* Defined as the fraction of true results (TP and TN) from all the activities considered.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

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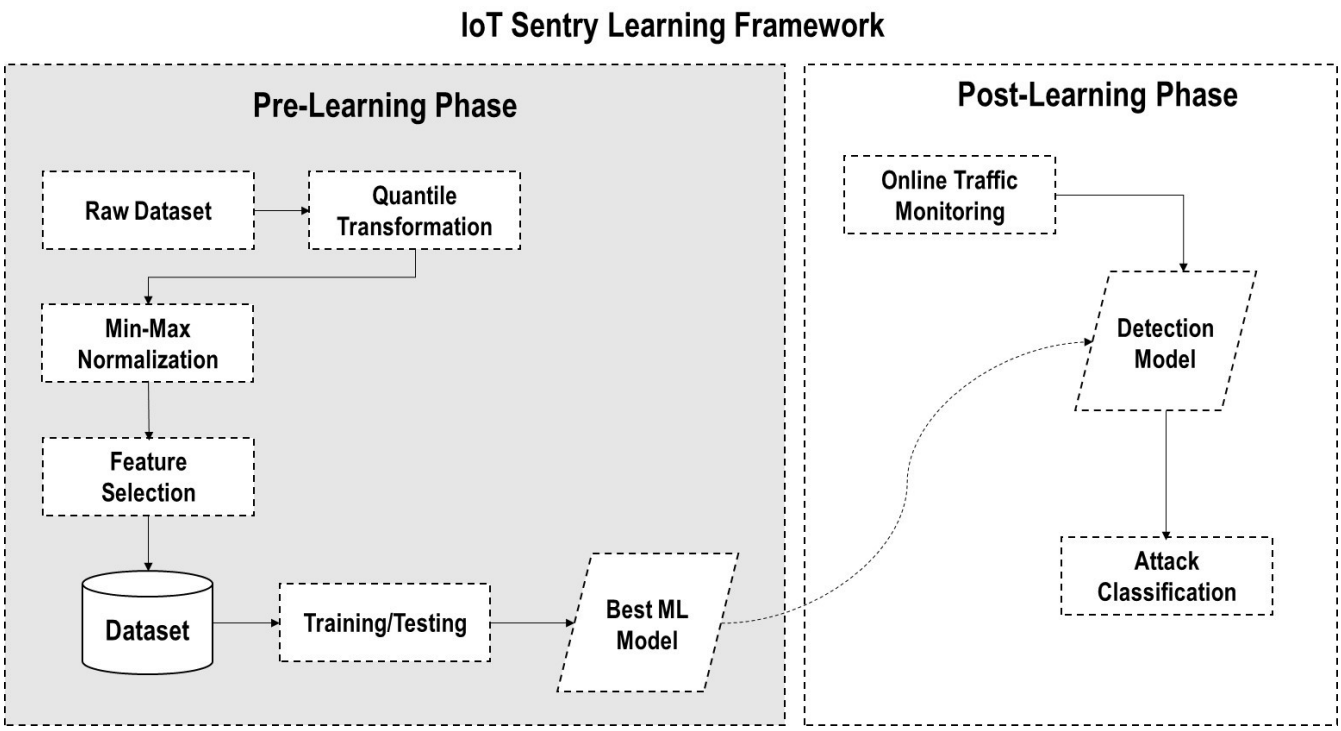

Fig. 4. IoT sentry learning framework.

TABLE VIII
CONFUSION MATRIX

		Predicted Class	
		Positive	Negative
*Actual Class	Positive	TP*	FN*
	Negative	FP*	TN*

*TP, FN, FP, TN represent True Positive, False Negative, False Positive and True Negative respectively.

- *Precision and Recall:* These metrics give further insights into classification performance. Precision is the ratio of correctly recognized intrusions relative to the predicted instances of intrusion activities. It is calculated using the equation:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall or True positive rate (TPR) or detection rate is the ratio of recognized intrusions to the total number of intrusion instances and calculated as follows:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

- *F-score:* In cases when only one accuracy metric is available for evaluation, F-score, also called F-measure or F1-score is preferred. It is the weighted average of recall and precision.

$$F_1 = 2 \times \frac{1}{\frac{1}{Recall} + \frac{1}{Precision}} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

IoT-Sentry deploys a binary classification algorithm to differentiate between benign and malicious traffic. To evaluate the classification performance, we analyze the overall detection rate achieved by IoT-Sentry and compare against results from some of the literature. All the detection results are obtained with 30% hold-out validation method on the IoT-CIDDS dataset.

Müller *et al.* [26] generated a prototype of attacks in C language for attacks as listed in Table II. Their prototype achieved detection rate of 91% with only single attacker in the network. Napiah *et al.* [25] proposed CHA-IDS based on analysis of 6LoWPAN header for feature extraction. CHA-IDS achieved 99% detection rate with 8 nodes and single attacker. The IDS by Canbalaban and Sen [31] utilized cross-layer features to detect attacks with different attacker scenarios on 50 nodes and achieved detection rate of 96.88% on average. Yavuz *et al.* [24] developed the IRAD dataset with traffic traces generated in the same way as ours and incorporated similar set of features for detecting hello flood and blackhole attacks. Thus, it becomes reasonable to directly compare the performance of these two attacks in [24] against ours. The advantage of IoT-Sentry over [24] is its simplicity, reduced training time and the need for fewer computational resources for implementation of ensemble learning on the 6LoWPAN border router. The Hello Flood and blackhole attacks' detection performance of IoT-Sentry compared with Yavuz *et al.* [24] is given in Table IX. The enhanced performance of IoT-Sentry is due to our additional features and deployment of ensemble learning model for intrusion detection. Besides, the implementation of deep learning with seven layered neural network in [24] incurs longer training time and higher computational resources for training and validation. On average, the attack detection rate of [24] was measured as 95.66%.

Figure 5 shows the detection rates for different proposals compared against IoT-Sentry. Though, the performance of CHA-IDS [25] is equivalent to ours, it has been tested only on

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TABLE IX
COMPARISON OF PERFORMANCE OF DIS/HELLO FLOOD ATTACK

	Blackhole			Hello Flood		
	Precision	Recall	F-score	Precision	Recall	F-score
IRAD	95	96	0.94	98.26	99.03	0.9862
IoT-Sentry	99.98	99.72	0.998	98	97	0.98

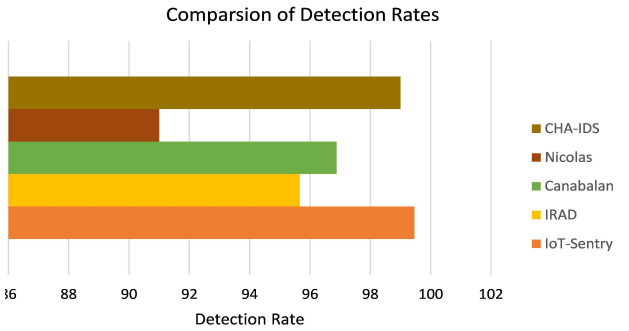

Fig. 5. Comparison of detection rates.

TABLE X
PERFORMANCE OF PROPOSED IDS

Attack Scenario	Accuracy	Precision	Recall	F1 Score
UDP Flooding	99.66	99.60	99.7	0.9968
SF	98.97	99.2	99.02	0.9911
BK	99.87	99.98	99.72	0.9985
HF	98.77	98.26	99.03	0.9864
ICMPv6 Flood	99.4	99.13	99.42	0.9927

eight nodes with single attacker. IoT-Sentry achieves significant improvement in detection rates (with 99.46%) compared to its contemporaries discussed above with similar scale of network.

Our proposed IDS, IoT-Sentry has been evaluated on five different IoT layer attacks on a large IoT network with up to 100 nodes. We generated our own dataset with multiple attacker scenarios given in Table IV and extracted features with our own prototype implementation in Python. To increase the accuracy of our detection model, we selected the significant features using Random Forest. Further, the ensemble learning model was used to training and with that we achieved average detection rates of 99.46%. We tested IoT-Sentry on datasets created from a number of different scenarios to ascertain the correctness of our model. These scenarios include 10, 20, 40 and 100 nodes with multiple attacker nodes as shown in Table IV. The results that we present are average of all tests. The prediction performance of our proposed model on different attacks are listed in Tables X.

VI. CONCLUSION AND FUTURE WORK

This work is baseline towards development of cross layer intrusion detection in IoT with a comprehensive dataset of diverse attacks against standardized IoT stack. Our study proposes the first cross layer IoT dataset with routing, network and transport layer attacks. Based on the dataset, we propose IoT-Sentry, capable of detecting five attacks. This work is a novel work for cross layer attack detection in IoT networks. IoT-Sentry is lightweight as it does not add any overhead on the constrained network and devices. Beginning from extracting candidate features and an initial random forest model, the feature normalization and selection results in specific detection model for each type of attack. Since we have obtained high accuracy, precision, recall and F-score for attacks, we can say that the IoT-Sentry can be generalized over other similar attacks and different network topologies also. We built our attack datasets with simulation in Contiki which perfectly emulates real sensors and standardized protocol implementations. The CIDDS dataset includes upto 100 nodes and multiple attackers which is a representational scale for real world IoT networks. We developed Random Forest Classifier model trained with CIDDS dataset with high accuracy and precision. We got metrics as good as 99.9%

For future work, we plan to extend the capabilities of IoT-Sentry concerning: (i) evaluation on real IoT hardware testbed running actual IoT applications, (ii) investigating the performance with dynamic IoT network, (iii) include diverse and simultaneous attacks in CIDDS dataset, (iv) improving the performance by investigating more features and comparing performance of IoT-Sentry with state of the art proposals, and (v) include multi-class classification with attacker identification in IoT-Sentry.

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